

Lesson 5-1: Determine the Area of Parallelograms and Rhombuses

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute 24×13 .

Answer: 312

$$24 \times 10 + 24 \times 3 = 240 + 72.$$

Prerequisite skill: Multiply two- and three-digit numbers

TASK 2 • FOUNDATION 2

A rectangle is 8 cm long and 6 cm wide. Find its area in square centimeters.

Answer: 48

Multiply length by width: $8 \times 6 = 48$.

Prerequisite skill: Find the area of a rectangle by multiplying length by width

TASK 3 • FOUNDATION 3

A parallelogram has base 10 cm, slanted side 7 cm, and perpendicular height 4 cm. Which number is the height?

Answer: 4

Height meets the base at a right angle; the slanted side is not the height.

Prerequisite skill: Tell a base from a slanted side and identify the perpendicular height

TASK 4 • FOUNDATION 4

A rectangle measures 3 meters by 5 meters. State its area with units.

Answer: 15 square meters

$3 \times 5 = 15$; area counts square units.

Prerequisite skill: Label an area answer in square units

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TASK 5 • QUICK CHECK A

Area of an 8 cm by 5 cm rectangle? In a parallelogram, is the height the slanted side?

Answer: Rectangle area = $8 \text{ cm} \times 5 \text{ cm} = 40 \text{ cm}^2$. Height question: No! The height must be perpendicular to the base, never the slanted side.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Area of a 9 cm by 7 cm rectangle? In a parallelogram, is the height the slanted side?

Answer: $9 \times 7 = 63 \text{ cm}^2$. No — the height must meet the base at a right angle; the slanted side is longer than the height.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Area of a 12 m by 4 m rectangle? A parallelogram has base 12 m, slant side 5 m, height 4 m. What is its area?

Answer: Rectangle: $12 \times 4 = 48 \text{ m}^2$. Parallelogram: $12 \times 4 = 48 \text{ m}^2$ — use the height 4, not the slant 5.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Two parallelograms both have base 10 cm and height 6 cm but different slant sides. Compare their areas.

Answer: Both are 60 cm^2 . The slant side does not affect area — only base and perpendicular height do.

Check the result against the situation and explain your reasoning.

Lesson 5-2: Determine the Area of Triangles

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $(14 \times 9) \div 2$.

Answer: 63

Halve 14 first to get 7, then multiply 7×9 .

Prerequisite skill: Multiply, then halve — or halve, then multiply

TASK 2 • FOUNDATION 2

Compute $1/2 \times 38$.

Answer: 19

Multiplying by one half is the same as dividing by 2.

Prerequisite skill: Multiply a whole number by $1/2$

TASK 3 • FOUNDATION 3

A parallelogram has base 12 cm and height 5 cm. Find its area in square centimeters.

Answer: 60

Area is base $\times$ perpendicular height: 12×5 .

Prerequisite skill: Find the area of a rectangle or parallelogram

TASK 4 • FOUNDATION 4

A triangle has base 8 cm, perpendicular height 6 cm, and slanted side 10 cm. Which length belongs in the height part of its area formula?

Answer: 6

Use the length perpendicular to the chosen base.

Prerequisite skill: Identify the height drawn perpendicular to the chosen base

Lesson 5-2: Determine the Area of Triangles

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Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

7×4 , then half? Area of a triangle with base 7 m and height 4 m?

Answer: $7 \times 4 = 28$, then half = 14. Triangle area: $A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 7 \text{ m} \times 4 \text{ m} = 14 \text{ m}^2$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

9×6 , then half? Area of a triangle with base 9 cm and height 6 cm?

Answer: $9 \times 6 = 54$, half is 27. Area = $\frac{1}{2} \times 9 \times 6 = 27 \text{ cm}^2$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

12×5 , then half? Area of a triangle with base 12 in and height 5 in?

Answer: $12 \times 5 = 60$, half is 30. Area = $\frac{1}{2} \times 12 \times 5 = 30 \text{ in}^2$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A triangle has area 36 cm^2 and base 12 cm. What is its height?

Answer: $36 = \frac{1}{2} \times 12 \times h$, so $36 = 6h$ and $h = 6 \text{ cm}$.

Check the result against the situation and explain your reasoning.

Lesson 5-3: Determine the Area of Trapezoids

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $(7 + 11) \times 4$.

Answer: 72

Add inside parentheses first: $18 \times 4 = 72$.

Prerequisite skill: Add the two numbers inside parentheses before multiplying

TASK 2 • FOUNDATION 2

Compute $1/2 \times 54$.

Answer: 27

Half of 54 is 27.

Prerequisite skill: Multiply by $1/2$

TASK 3 • FOUNDATION 3

Evaluate $1/2 \times (6 + 10) \times 5$.

Answer: 40

First $6 + 10 = 16$, then half of 16 is 8, then $8 \times 5 = 40$.

Prerequisite skill: Follow order of operations inside a formula

TASK 4 • FOUNDATION 4

A shape is made of a rectangle of area 24 and two triangles of area 6 each, without overlap. Find the total area.

Answer: 36

Add the three nonoverlapping areas: $24 + 6 + 6$.

Prerequisite skill: Split a figure into a rectangle plus triangles

Lesson 5-3: Determine the Area of Trapezoids

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$\frac{1}{2} \times (6 + 10) \times 4 = ?$ Do the parentheses first.

Answer: Parentheses first: $(6 + 10) = 16$. Area = $\frac{1}{2} \times 16 \times 4 = 8 \times 4 = 32$ square units.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$\frac{1}{2} \times (8 + 12) \times 5 = ?$ Do the parentheses first.

Answer: $(8 + 12) = 20$. Then $\frac{1}{2} \times 20 \times 5 = 10 \times 5 = 50$ square units.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$\frac{1}{2} \times (7 + 9) \times 6 = ?$ Do the parentheses first.

Answer: $(7 + 9) = 16$. Then $\frac{1}{2} \times 16 \times 6 = 8 \times 6 = 48$ square units.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A trapezoid has bases 6 and 10 and area 48. What is its height?

Answer: $48 = \frac{1}{2}(6 + 10)h = 8h$, so $h = 6$ units.

Check the result against the situation and explain your reasoning.

Lesson 5-4: Apply Area Concepts to Solve Problems

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Find the area of a triangle with base 10 cm and height 8 cm, in square centimeters.

Answer: 40

Half of 10×8 is 40.

Prerequisite skill: Find the area of rectangles, triangles, and trapezoids on demand

TASK 2 • FOUNDATION 2

A rectangle has area 72 square meters. A 15-square-meter section is removed. What area remains?

Answer: 57

Subtract the removed area: $72 - 15 = 57$.

Prerequisite skill: Add and subtract areas

TASK 3 • FOUNDATION 3

A total side length is 14 cm. One part is 9 cm. How long is the other part?

Answer: 5

Subtract known part from total: $14 - 9 = 5$.

Prerequisite skill: Find a missing side length from the other labeled sides

TASK 4 • FOUNDATION 4

Six identical tiles each cover 12 square centimeters. What is the total area?

Answer: 72

Multiply area per tile by number of tiles: 12×6 .

Prerequisite skill: Multiply one part's area by the number of identical parts

Lesson 5-4: Apply Area Concepts to Solve Problems

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

A 10 by 6 rectangle has a 3 by 2 rectangle cut out. What area is left?

**Answer: Large rectangle area: $10 \times 6 = 60$.
Cutout rectangle area: $3 \times 2 = 6$. Area remaining: $60 - 6 = 54$ square units.**

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

A 12 by 8 rectangle has a 4 by 3 rectangle cut out. What area is left?

**Answer: $12 \times 8 = 96$. Cut-out: $4 \times 3 = 12$.
Remaining: $96 - 12 = 84$ square units.**

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

A 9 by 9 square has a triangle of base 6 and height 4 cut out. What area is left?

**Answer: Square: 81. Triangle: $\frac{1}{2} \times 6 \times 4 = 12$.
Remaining: $81 - 12 = 69$ square units.**

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A rectangular garden 15 m by 10 m has a 3 m by 3 m square shed inside. How much garden is left to plant?

Answer: $150 - 9 = 141$ m².

Check the result against the situation and explain your reasoning.

Lesson 5-5: Determine the Volume of Rectangular Prisms

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $4 \times 7 \times 5$.

Answer: 140

Group $4 \times 5 = 20$, then $20 \times 7 = 140$.

Prerequisite skill: Multiply three factors in any order

TASK 2 • FOUNDATION 2

Compute $1\frac{1}{2} \times \frac{2}{3}$.

Answer: 1

Convert to $\frac{3}{2}$, then multiply $\frac{3}{2} \times \frac{2}{3} = \frac{6}{6}$.

Prerequisite skill: Multiply fractions and mixed numbers

TASK 3 • FOUNDATION 3

A rectangular base is 9 cm by 4 cm. Find its area in square centimeters.

Answer: 36

Multiply $9 \times 4 = 36$ square centimeters.

Prerequisite skill: Find the area of a rectangular base

TASK 4 • FOUNDATION 4

A prism measures 2 cm by 3 cm by 5 cm. State its volume with units.

Answer: 30 cubic centimeters

Multiply $2 \times 3 \times 5 = 30$; volume counts cubic units.

Prerequisite skill: Label a volume answer in cubic units

Lesson 5-5: Determine the Volume of Rectangular Prisms

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$4 \times 3 \times 2$? $\frac{1}{2} \times \frac{1}{3}$? Volume of a 4 by 3 by 2 prism?

Answer: $4 \times 3 \times 2 = 24$. $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$. Volume = $4 \times 3 \times 2 = 24$ cubic units.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$5 \times 4 \times 3$? $\frac{1}{2} \times \frac{1}{4}$? Volume of a 5 by 4 by 3 prism?

Answer: $5 \times 4 \times 3 = 60$. $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$. Volume = 60 cubic units.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$6 \times 2 \times 4$? $\frac{2}{3} \times \frac{1}{2}$? Volume of a 6 by 2 by 4 prism?

Answer: $6 \times 2 \times 4 = 48$. $\frac{2}{3} \times \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$. Volume = 48 cubic units.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A box is $3\frac{1}{2}$ in by 2 in by 4 in. What is its volume?

Answer: $3.5 \times 2 = 7$, and $7 \times 4 = 28 \text{ in}^3$.

Check the result against the situation and explain your reasoning.

Lesson 5-6: Represent Three-Dimensional Figures in Two Dimensions

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

How many faces, edges, and vertices does a rectangular prism have?

Answer: 6 faces, 12 edges, 8 vertices

Count the six flat surfaces, twelve joining segments, and eight corners.

Prerequisite skill: Name the faces, edges, and vertices of a prism and a pyramid

TASK 2 • FOUNDATION 2

How many faces does a square pyramid have, including its base?

Answer: 5

One square base plus four triangular faces gives five faces.

Prerequisite skill: Count the faces of a solid

TASK 3 • FOUNDATION 3

A net has a rectangular face 3 cm by 7 cm. Find that face's area in square centimeters.

Answer: 21

Multiply 3×7 ; count this face once.

Prerequisite skill: Find the area of each rectangle and triangle inside a net

TASK 4 • FOUNDATION 4

A prism has face pairs sized 3 by 4, 3 by 5, and 4 by 5. How many faces are 3 by 4?

Answer: 2

Each rectangular face has a congruent opposite face.

Prerequisite skill: Match congruent faces in pairs

Lesson 5-6: Represent Three-Dimensional Figures in Two Dimensions

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

How many faces does a rectangular prism have?
How many matching pairs is that?

Answer: A rectangular prism has 6 faces. That forms 3 matching pairs of congruent opposite faces (top/bottom, front/back, left/right).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

How many faces does a triangular prism have?
How many are triangles?

Answer: 5 faces — 2 triangular bases and 3 rectangular sides.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

How many faces does a square pyramid have?
How many are triangles?

Answer: 5 faces — 1 square base and 4 triangles.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A net has two congruent right triangles with side lengths 3, 4, and 5 units, and three rectangles measuring 3 by 10, 4 by 10, and 5 by 10 units. The triangle legs of lengths 3 and 4 are perpendicular. What solid does the net form, and what is its surface area?

Answer: A right triangular prism. The two triangular bases have total area $2 \times (1/2 \times 3 \times 4) = 12$ square units. The rectangles have total area $30 + 40 + 50 = 120$ square units. Surface area = $12 + 120 = 132$ square units.

Check the result against the situation and explain your reasoning.

Lesson 5-7: Determine Surface Area of Prisms

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Add the face areas $12 + 12 + 15 + 15 + 20 + 20$.

Answer: 94

Double each pair: $24 + 30 + 40 = 94$.

Prerequisite skill: Add a list of five or six areas

TASK 2 • FOUNDATION 2

Find the area of a triangle with base 6 cm and height 9 cm, in square centimeters.

Answer: 27

Half of 6×9 is 27.

Prerequisite skill: Find the area of rectangles and triangles quickly

TASK 3 • FOUNDATION 3

In a rectangular prism with dimensions 2, 3, 5, list the three different face areas.

Answer: 6, 10, 15

Compute 2×3 , 2×5 , and 3×5 ; each appears twice.

Prerequisite skill: Read a net and pair up matching faces

TASK 4 • FOUNDATION 4

One face has area 18 square centimeters. What is the combined area of it and its matching opposite face?

Answer: 36

Double the face area: 2×18 .

Prerequisite skill: Double one face's area to cover a matching pair

Lesson 5-7: Determine Surface Area of Prisms

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Faces measure 12, 12, 20, 20, 15, 15 square units — what is the surface area?

Answer: Surface Area = $12 + 12 + 20 + 20 + 15 + 15 = 24 + 40 + 30 = 94$ square units.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Faces measure 18, 18, 24, 24, 30, 30 square units — what is the surface area?

Answer: $36 + 48 + 60 = 144$ square units.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

A prism has faces 10, 10, 14, 14, 35, 35 square units — what is the surface area?

Answer: $20 + 28 + 70 = 118$ square units.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A cube has surface area 150 cm^2 . What is the length of one edge?

Answer: Each of 6 faces is $150 \div 6 = 25 \text{ cm}^2$, so an edge is 5 cm.

Check the result against the situation and explain your reasoning.

Lesson 5-8: Determine Surface Area of Pyramids

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Find the area of a triangle with base 8 cm and perpendicular height 7 cm, in square centimeters.

Answer: 28

Half of $8 \times 7 = 56$ is 28.

Prerequisite skill: Find the area of a triangle

TASK 2 • FOUNDATION 2

Find the area of a square base with side 6 cm, in square centimeters.

Answer: 36

A square has equal length and width: 6×6 .

Prerequisite skill: Find the area of a square or rectangular base

TASK 3 • FOUNDATION 3

Four congruent triangular faces each have area 21 square centimeters. Find their combined area.

Answer: 84

Multiply one face area by four: 21×4 .

Prerequisite skill: Multiply one triangular face's area by the number of identical faces

TASK 4 • FOUNDATION 4

A pyramid's vertical height is 9 cm. The altitude drawn within a triangular face is 10 cm. Which height finds that face's area?

Answer: 10

Use the altitude lying in the triangular face, called the slant height.

Prerequisite skill: Tell slant height from vertical height on a net or drawing

Lesson 5-8: Determine Surface Area of Pyramids

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

One triangular face is 24 square inches. With four identical faces, what is the lateral area?

Answer: Lateral area = $4 \times 24 = 96$ square inches.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

One triangular face is 30 square inches. With four identical faces, what is the lateral area?

Answer: $4 \times 30 = 120$ in².

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

A square pyramid has four congruent triangular faces, each with area 18 cm², and a square base with area 36 cm². What is its total surface area?

Answer: Lateral: $4 \times 18 = 72$. Plus base: $72 + 36 = 108$ cm².

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A square pyramid has base edge 10 cm and slant height 12 cm. Find the total surface area.

Answer: One triangle: $\frac{1}{2} \times 10 \times 12 = 60$. Four triangles: 240. Base: $10 \times 10 = 100$. Total: 340 cm².

Check the result against the situation and explain your reasoning.